



MULTIMEDIA UNIVERSITY OF KENYA

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

UNIVERSITY EXAMINATIONS 2020/2021

SECOND YEAR FIRST SEMESTER SUPPLEMENTARY/SPECIAL EXAMINATIONS

FOR THE DEGREE OF

BACHELOR OF SCIENCE IN COMPUTER SCIENCE/ BACHELOR OF SCIENCE IN

INFORMATION TECHNOLOGY/ BACHELOR OF SCIENCE IN SOFTWARE

ENGINEERING

CCS 2217/ SMA 2103: PROBABILITY AND STATISTICS I/ PROBABILITY AND
STATISTICS

DATE: 16TH JUNE 2022

TIME: 2 HOURS

INSTRUCTIONS:

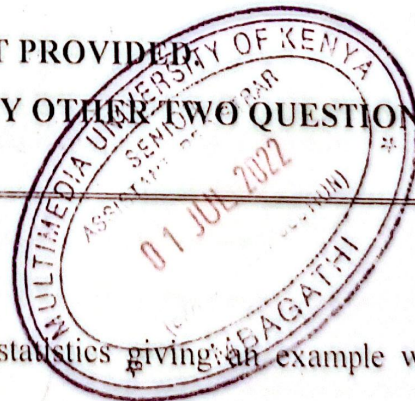
ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.
ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS.

QUESTION ONE (THIRTY MARKS)

(a) Differentiate between the following terms as used in statistics giving an example where possible.

- (i) Census and sample survey.
- (ii) Qualitative and Quantitative variable
- (iii) Primary and secondary data
- (iv) Stratified and simple random sampling
- (v) skewness and kurtosis

(10 marks)



(b) The following are examples of scales used in the measurement of statistical data. Classify them into the following measurement scales: nominal, ordinal, interval and ratio:

- sex of staff as male and female;
- marks scored by students in an examination;
- daily temperature in degrees Celsius at the campus whether station;
- grades scored by students in an examination;
- age in years of patients who attended a clinic;
- numbers on the back of players in a football match;
- level of customer satisfaction as low, moderate or high;
- longitude of a place on earth surface.

(4 marks)

(b) Explain in words each of the following terms as used in Statistics:

- (i) mean;
- (ii) median.

(4 marks)

(c) The data given below represents the frequency distribution of pairs of shoes and the various shoe sizes purchased by 60 customers during the year 2020.

Size of Shoes	4	5	6	7	8	9	10
No. of Pairs	1	11	5	27	4	10	2

Determine each of the following measures about the distribution of the sizes of the shoes sold:

- (i) Median;
- (ii) Mean;
- (iii) Standard deviation.

(7 marks)

(e) A box contains 8 green apples and 4 red apples which are identical in all aspects except colour. A customer picks 2 apples at random from the box one at a time without replacement.

- (i) Present this information using a probability tree diagram.
- (ii) Hence determine the probability that all the apples picked are of different colours.

(3 marks)

QUESTION TWO (20 MARKS)

(2 marks)

The table below shows the age distribution of 100 employees working at the headquarters of a county government.

Age (years)	20 - 24	25 - 29	30 - 34	35 - 39	40 - 44	45 - 49
No. of employees	15	5	30	20	10	20

(4 marks)

- (a) Estimate the standard deviation of the age of the employees.
- (b) Due to continuous burden of the huge wage bill, it is desired to reduce the present number of employees according to the following scheme:
- To retrain the first 15% from the lower age group.
 - To retain the next 45% in their current workstations.
 - To transfer the next 30% to the national government.
 - To retrench the 10% from the highest age group.

Determine the age limits of the persons retained and those to be transferred to the national government. (8 marks)

- (c) Suppose the scheme in (b) is revised based on the following criteria:
- To retrain the employees whose age is below 27 years;
 - To retain the employees whose age is between 27 and 32 years at their current work-stations.
 - To transfer the employees whose age is between 32 and 42 years to the national government;
 - To retrench the employees whose age is above 42 years.

Determine the proportion of the employees to be retained and those to be transferred to the national government.

(8 marks)

QUESTION THREE (20 MARKS)

The Ministry of Lands and Housing is interested in investigating the relationship between the monthly rent of a house and the distance from the nearest tarmac road in Kajiado County. A survey was carried out in which a random sample of 12 typical one bed-roomed residential houses in Kajiado County was taken. The data relating the distance from the nearest tarmac road in hundred metres and the monthly rent in thousand Kenya shillings is as shown in the table below.

House	A	B	C	D	E	F	G	H	J
Distance	16	8	25	18	20	30	24	12	15

Rent	10	16	13	11	20	12	12	25	10
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- (a) (i) Compute the Pearson's product moment co-efficient of correlation between the distance from the tarmac road and the monthly rent paid for the house. (9 marks)
- (ii) Interpret the result obtained in (a) (i). (2 marks)
- (b) (i) Determine the least squares regression line of the monthly rent paid on the distance from the nearest tarmac road. (4 marks)
- (ii) Interpret the regression line obtained in (b) (i). (3 marks)
- (iii) Using the regression line obtained in (b)(i), estimate by calculation the monthly rent for a typical two bed-roomed house which is located 2000 metres away from the nearest tarmac road. (2 marks)

QUESTION FOUR (20 MARKS)

- (a) Consider a simple linear regression model $y_i = a + bx_i + e_i$. Using the least squares technique, show that the regression intercept a and the regression co-efficient b in the regression model $\hat{y}_i = a + bx_i$ is given by: (11 marks)

$$b = \frac{\sum xy - n \bar{x} \bar{y}}{\sum x^2 - n \bar{x}^2} \quad \text{and} \quad a = \bar{y} - b \bar{x}$$

- (b) A survey on ICT usage conducted by the Ministry of Communication to study the ownership of ICT gadgets among residents in the County of Kajiado. A random sample of 450 residents was taken in the County and the findings on the ownership of a desktop computer, laptop computer and an i-pad were as shown below.
- 220 residents own a desktop computer
 - 216 residents own a laptop computer
 - 180 residents own an i-pad
 - 80 residents own a desktop computer and a laptop computer
 - 88 residents own a desktop computer and an i-pad
 - 72 residents own a laptop computer and an i-pad
 - 42 residents did not own any of the three electronic gadgets
- (i) Present this information in a Venn diagram. (5 marks)
- (ii) Determine the number of residents in the study who own:
- I. exactly two of the ICT gadgets;
 - II. a desktop computer or laptop computer but not an i-pad. (4 marks)

QUESTION FIVE (20 MARKS)

(a) Explain the term *conditional probability* as used in Statistics.

(2 marks)

(b) There is an outbreak of a given mysterious disease in a certain area. This disease has a known symptom. However, doctors realized that not all people who have the disease display the symptom and not all those who display the symptom have the disease. After conducting a medical survey among a random sample of 240 residents in the area, the findings were as follows:

- 96 patients who had the disease displayed the symptom;
- 64 patients who had the disease did not display the symptom;
- 32 patients who displayed the symptom did not have the disease;
- 48 patients did not have the disease and did not display the symptom;

Determine the following probabilities:

- (i) A patient who has the disease is selected at random from the area, determine the probability that he displays the symptom. (3 marks)
- (ii) A patient who displays the symptom is selected at random from the area, determine the probability that he does not have the disease. (3 marks)

(c) In a certain mechanical machine, the following actions are possible; the top can blow or not; the spiggots can go up or go down or remain still; and the cum-lever can wriggle or thump. The top blows with a probability of $\frac{1}{5}$. The spiggots go up with a probability of $\frac{1}{2}$ if the top blows; otherwise they go up or go down or remain still with equal probabilities. The cum-lever wriggles if the spiggots go up, otherwise it wriggles with a probability of $\frac{1}{6}$. The machine is robust and works provided the cum-lever is thumping.

- (i) Present this information in a probability tree diagram;
- (ii) Determine the probability that the machine works;
- (iii) Determine the probability that the top blows and the cum lever wriggles;
- (iv) If the machine breaks down, determine the probability that the top is blowing. (12 marks)



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UNIVERSITY EXAMINATIONS 2019/2010
THIRD YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN COMPUTER SCIENCE AND BACHELOR OF SCIENCE IN
SOFTWARE ENGINEERING

CCS 2217: PROBABILITY AND STATISTICS I

DATE: 10TH DECEMBER 2019

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (THIRTY MARKS)

- a) The height (x cm) of a group of 100 students were recorded. The summary data is given as

$$\sum fx = 17100 \text{ and } \sum fx^2 = 2926225$$

Work out the standard deviation.

(4 marks)

- b) The weekly income, x , of 100 women workers was recorded. The data were coded using

$\frac{x - 90}{100}$ and the following summations were obtained.

$$\sum y = 131 \text{ and } \sum y^2 = 176.84$$

Work out an estimate for:

- The mean of the actual women workers weekly income. (3 marks)
 - The standard deviation of the actual women workers' weekly income. (3 marks)
- c) In a study of how the mass of USB manufactured by a random sample of 11 companies was examined for a particular two weeks period. The mass of an USB, X grams, for the 11 companies were

- 20 15 10 30 33 40 5 11 13 20 25 42 31
- Find the median and quartiles for the data. (6 marks)
 - A value that is greater than $Q_3 + 1.5 \times (Q_3 - Q_1)$ or smaller than $Q_1 - 1.5 \times (Q_3 - Q_1)$ is defined as an outlier. Find the outlier in this data. (3 marks)
 - Eight children had their IQ measured and then took a general knowledge test. Their IQ, x , and their marks, y , for test were summarized as follows: (5 marks)

$$\sum x = 973, \sum x^2 = 120123, \sum y = 490, \sum y^2 = 33000, \sum xy = 61595$$

- Calculate the product moment correlation coefficient. (4 marks)
 - Describe and product the correlation coefficient between IQ and general knowledge. (2 marks)
- e) Of the households in the Kenya, 40% have a plasma TV and 50% have a laptop computer. There are 25% of households that have both a plasma TV and a laptop. Find the probability that a household chosen at random has either a plasma TV or a laptop computer but not both. (5 marks)

QUESTION TWO (TWENTY MARKS)

- a) Samantha and Shelley talk to one another once a week on the telephone. The table below shows the number of minutes (to nearest minute) spent in conversion on each occasion.

Time (to nearest minute)	5-9	10-14	15-19	20-24	25-29	30-34
No of conversions	2	9	20	13	8	3

- Two of the conversions were chosen at random. Find the probability that both were longer than 24.5 minutes. (2 marks)
 - Calculate an estimate of the mean time spent on their conversions. (4 marks)
 - During the following 25 weeks they monitored their weekly conversions and found out that at the end of 80 weeks their overall mean length of conversions was 21 minutes. Find the mean time spent in the conversions during these 25 weeks. Comment on these two mean values. (4 marks)
- b) The weekly wages (to the nearest £) of the production line workers in a small factory are shown in the table below:

Weekly wage £	175-225	226-300	301-350	351-400	401-500
Number of workers	4	8	18	28	7

- Calculate the estimate of the mode wage. (3 marks)
- Calculate the estimate of the mean wage. (4 marks)
- Calculate the estimate of the median wage.. (3 marks)

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QUESTION THREE (TWENTY MARKS)

- a) A group of lecturers recorded the time spent on the first meeting with a random sample of 120 students in the mentorship programme. The times, to the nearest minute, are summarized in the table below.

Time	10-15	15-20	20-25	25-30	30-35	35-45	45-80
No of students	2	5	17	38	29	25	4

- Calculate estimates of the mean and variance. (5 marks)
 - Obtain estimates of the median and quartiles of the times spent with customers. (6 marks)
 - Compute the Pearson's second coefficient of skewness $\frac{3(\text{mean}-\text{median})}{\text{standard deviation}}$ and describe the skewness of these data. (2 marks)
 - The lecturers are undecided whether to use the median and quartiles, or the mean and standard deviation to summarize these data. State, giving a reason, which you would recommend them to use. (2 marks)
- b) Data below represents marks scored by a group of students.

Class	40-45	45-50	50-55	55-60	60-65	65-70
Frequency	4	13	14	12	5	2

Compute the moment coefficient of kurtosis and comment on it, (5 marks)

QUESTION FOUR (TWENTY MARKS)

In a training scheme for young people, the average time taken for each age group to reach a certain level of proficiency was measured. The data are shown in the table below:

Age x (years)	16	17	18	19	20	21	22	23	24	25
Average time y (hours)	12	11	10	9	11	8	9	7	6	8

- Calculate the product moment correlation coefficient (r). (6 marks)
- Describe and interpret the relationship between average time and age. (3 marks)
- Estimate the least square regression equation of average time against age. (6 marks)

- d) Use your results in (iii) to find the value of the average time for young person aged 20 years and also find the age for a person who's average time is $10\frac{1}{2}$ hours (5 marks)

QUESTION TWO (TWENTY MARKS)

- a) On a randomly chosen day the probability that Bill travels to college by bus, by bicycle or on foot is and respectively. The probability of being late when using these methods of travel is and respectively. (2 marks)
- i). Draw a tree diagram to represent this information. (2 marks)
 - ii). Find the probability that on a randomly selected day Bill travels by foot and is late. (2 marks)
 - iii). Find the probability that on a randomly selected day Bill is late. (4 marks)
 - iv). Given that Bill is late, find the probability that he did not travel on foot. (3 marks)
- b) One objective of a computer game is to collect keys. There are three stages to the game. The probability of collecting a key at the first stage is , at the second stage is , and at the third is . (3 marks)
- i). Calculate the probability of collecting all 3 keys. (3 marks)
 - ii). Calculate the probability of collecting exactly one key in a game. (4 marks)
 - iii). Calculate the probability that keys are not collected on at least 2 successive stages in a game.



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UNIVERSITY EXAMINATIONS 2019/2010
THIRD YEAR FIRST SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

SMA 2103: PROBABILITY AND STATISTICS I

DATE: 10TH DECEMBER 2019

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE [COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (TWENTY MARKS)

Classify the following variable as discrete or continuous or categorical. Also identify the level of scale of measurement as either nominal or ordinal or interval or ratio.

- The names of students in the BIT Y2S1 lecture attendance sheet. (2 marks)
- The number of student in a lecture room. (2 marks)
- Lifetime in hours of cell phone batteries. (2 marks)

Meryl records the mass (in kg) of mobile phones displayed in MMU museum. The result is shown in the below.

500 700 400 300 900 700 700

- Calculate the mean, the mode and the median mass of the mobile of these phones (3 marks)
- A week after, Meryl was reported to that another old mobile phone weighing 650 grams was donated to the MMU museum. write down the effect of this will have on the mean. Mode and variance. (2 marks)

The noise levels at 30 locations near an outdoor church rallies venue were measured to the nearest decibel. The data collected is recorded as in the table below:

Noise (decibels)	65-69	70-74	75-79	80-84	85-89	90-94	95-99
No of computers	1	4	6	6	8	4	1

M-D

- i). Estimate the mean noise level. (3 marks)
 ii). A noise level above 82 decibels was considered unacceptable. Estimate the number of locations that had unacceptable noise level. (2 marks)
 d) The ages, a years, and heights, h cm, of seven members of a team were recorded. The data was summarized as follows:

$$\sum a = 115, \sum a^2 = 1899, S_{hh} = 571.4, S_{ah} = 72.1$$

- i). Find the value of the product moment correlation coefficient between a and h . (3 marks)
 ii). Describe and interpret the correlation between the age and height of these seven people based on these data. (2 marks)
 e) The events A and B are independent with $P(A) = \frac{1}{4}$ and $P(A \cup B) = \frac{1}{5}$, find:
 i). $P(A \cap B)$. (3 marks)
 ii). $P(A \cap B^c)$. (1 mark)
 iii). $P(A^c \cap B^c)$. (1 mark)
 f) The random variable X has the probability function given by:

$$P(X = x) = \begin{cases} \frac{x-1}{10}, & x = 1, 2, 3, 4, 5. \\ 0, & \text{otherwise} \end{cases}$$

Find:

- i). $P(2 < X \leq 4)$ (2 marks)
 ii). $E(X)$. (2 marks)

QUESTION TWO (TWENTY MARKS)

- a) The weekly wages (to the nearest ksh00) of the production line workers in a small factory is shown in the table below

Weekly wages (to nearest ksh00)	175-225	226-300	301-350	351-400	401-500
No of workers	4	8	18	28	7

Calculate:

- i). The mode (2 marks)
 ii). The median (3 marks)
 iii). The mean (3 marks)
 iv). The standard deviation (2 marks)
 v). Compute the Pearson's second coefficient of skewness $\frac{3(\text{mean} - \text{median})}{\text{standard deviation}}$ and describe the skewness of these data. (2 marks)
 b) A computer retailer has two shops in Nairobi city. Shop A and shop B employs 15 and 22 people respectively. The mean and the standard deviation number of days of sickness in a year taken by employees is 4.6 and 2.1 in A and 6.5 and 2.4 days in shop B.
 i). Calculate the mean number of days of sickness taken per year by all 37 employees.

- ii). Calculate the standard deviation number of days of sickness taken per year by all 37 employees. (3 marks)

(5 marks)

QUESTION THREE (TWENTY MARKS)

As part of the study of charity shops in a small market town, X and Y , were each asked to provide details of its takings on 12 randomly selected days. The table shows, for each of the 12 days, the day's takings, £ x , of charity shop X and the day's takings, £ y , of charity shop Y .

Day	A	B	C	D	E	F	G	H	I	J	K	L
x	46	57	39	116	62	77	41	61	15	53	68	61
y	78	102	66	214	98	72	98	134	21	67	95	8

- (i) Calculate the value of the product moment correlation coefficient between X and Y . (5 marks)

- (ii) Interpret your value in the context of this question. (2 marks)

Construct the scatter diagram for these data. (5 marks)

The investigator realized subsequently that one of the 12 selected days was a particularly popular town market day and another was a day on which the weather was extremely severe.

Identify **each** of these days giving a reason for each choice. (3 marks)

Removing the two days described in part (c) from the data gives the following information.

$$S_{xx} = 1292.5, \quad S_{yy} = 3850.1, \quad S_{xy} = 407.5$$

- i). Use this information to recalculate the value of the product moment correlation coefficient between X and Y . (2 marks)

- ii). Hence revise, as necessary, your interpretation in part (a)(ii) (3 marks)

QUESTION FOUR (TWENTY MARKS)

A bag contains 24 electrical components of which four are known to be defective. Two items are taken at random from the bag. Find the probability of selecting:

- i). Two defective components if the selection is done with replacement. (2 marks)

- ii). Two defective components if the selection is done with replacement. (2 marks)

- iii). One defective component and one fully functioning component if the selection is done without replacement (3 marks)

Two events J and K are such that $P(J \cup K) = 0.5$, $P(J^c \cap K) = 0.2$ and $P(J \cap K^c) = 0.25$

- i). Draw a Venn diagram to represent the events J and K and the sample space S . (2 marks)

Hence find

(1 mark)

- ii). $P(J)$.

(1 mark)

- iii). $P(K)$.

iv). $P(J/K)$.

- c) A garage sells three types of fuel, U95, U98 and diesel. In a survey of 200 motorists buying fuel, 80 are females and the rest are males. Of the 90 motorists buying U95 fuel, 50 were female and the 70 motorists buying diesel, 60 were male. A motorist does not buy more than one type of fuel. Garage records indicates that 10% of the motorists buying U95 fuel, 30% of the motorists buying U98 fuel and 40% of the motorists buying diesel have their car serviced by the garage. A motorist is chosen at random. Given the motorist has his car serviced by the garage, find the probability that the motorist buys diesel fuel

(2 marks)
(7 marks)

QUESTION FIVE (TWENTY MARKS)

- a) When parking a car in a downtown parking lot, drivers pay according to the number of hours. The probability distribution for the number of hours a car is parked has been given below:

x	1	2	3	4	5	6	7	8
$P(X = x)$	0.24	0.18	0.13	0.10	0.07	0.04	0.04	0.20

- i). What is the probability that a car will be parked in this parking lot for at least 6 hours? (3 marks)
 ii). What is the expected number of hours a given car will be parked in the parking lot? (3 marks)
 iii). If revenue paid earns a profit, in ksh, given by

$$Y = 5X + 2$$

Find $E(Y)$ and $\text{var}(Y)$

- b) The discrete random variable X has p.d.f given by:

(4 marks)

$$f(x) = P(X = x) = \begin{cases} a(3-x), & x = 0, 1, 2 \\ b, & x = 3 \\ 0, & \text{otherwise} \end{cases}$$

- i). Given that $E(X) = 1.6$ find the value of a and b . (3 marks)
 ii). Find the value of $P(0.5 < X < 3)$. (2 marks)
 iii). Find $E(3X - 2)$. (2 marks)
 iv). Find $\text{var}(3X - 2)$. (3 marks)



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FIRST YEAR SECOND SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR SCIENCE IN INDUSTRIAL CHEMISTRY AND BACHELOR SCIENCE
IN ANALYTICAL CHEMISTRY

MAT 2103: PROBABILITY AND STATISTICS I

DATE: 29TH JULY 2021

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN ANSWER BOOKLET PROVIDED.

ANSWER QUESTION ONE[COMPULSORY] AND ANY OTHER TWO QUESTIONS

QUESTION ONE (THIRTY MARKS)

a) Explain the following as used in statistics;

(i) Descriptive Statistics

(1 Mark)

(ii) Sampling

(1 Mark)

(iii) Regression

(2 Mark)

(iv) Correlation

(1 Mark)

(5 Marks)

b) Explain five uses of statistics in your area of study

c) The following is some data (in '00) recorded by Mr. Nyamweya, a Human Resource Officer at a Kisii hotel, in a period of 20 days;

17.5 13.7 11.9 15.2 15.4 17.3 20.0 18.9 19.1 19.7

16.5 14.3 12.1 14.4 20.7 19.5 17.6 19.3 19.2 18.8

Calculate;

- (i) Mean (2 Marks)
- (ii) Standard deviation (3 Marks)
- d) Describe how you can determine skewness in data (5 Marks)
- e) The coefficient of variation of wages of male workers is 55% and that of female workers is 70%. The standard deviation of wages for male workers is 22 and that for female workers is 15.4. Calculate the overall mean wage off all workers given that 80% of the workers are male. (6 Marks)
- f) A box contains 20 DVDs, 4 of which are defective. If two DVDs are selected at random without replacement, what is the probability that both are defective? (4marks)

QUESTION TWO (TWENTY MARKS)

- a) Explain the importance of data presentation to a Human Resource Officer (3 Marks)
- b) Explain the difference between a bar chart and a histogram (6 Marks)
- c) Consider the following data on daily sale of mangoes (in '00 KShs.) obtained from a retailer at Gachororo market, Juja;

19.5 37.2 19.1 18.5 31.6 34.6 37.2 22.5 24.7 17.3
 33.4 16.8 22.6 24.6 21.2 29.8 34.1 13.4 17.8 34.4
 33.2 14.7 35.7 40.2 32.8 21.9 20.0 36.5 21.8 17.5
 19.5 22.5 33.1 38.2 31.7 23.5 18.4 24.7 22.8 36.1

- (i) Using an appropriate class interval, present this data in a frequency distribution table (7 Marks)
- (ii) From the frequency table, calculate the mean and variance (4 Marks)

QUESTION THREE (TWENTY MARKS)

A research studies the relationship between overheads based on the level of production. The data are collected as follows:

Overheads	56	42	36	47	49	42	60	72	63	55
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Production units	147	125	118	128	145	140	155	160	149	150
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- a) Calculate the correlation coefficient between overheads and production and comment on its value. (6 Marks)
- b) Calculate and state the regression model of production units an overhead. (10 Marks)
- c) Predict the overheads when 45 units are produced. (4 Marks)

QUESTION FOUR (TWENTY MARKS)

years	45	65	34	26	45	25	47	67	56
years	65	35	65	25	67	55	78	25	77

- a) Required;
- i) Find the correlation coefficient between the students (6 marks)
- ii) Comment on the outcome of the correlation (2 marks)
- iii) Find the spearman's rank correlation coefficient (5 marks)
- b) Discuss the various sampling techniques (7 marks)

QUESTION FIVE (TWENTY MARKS)

- a) A sample 10 students from a class was picked at random. Their performance in Advanced Accounting I was as follows:

92	58	61	82	60	53	70	84	77	72
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Required:

- (i) Harmonic mean (2 Marks)
- (ii) Arithmetic mean (2 Marks)
- (iii) 70th percentile (2 Marks)
- (iv) Geometric mean (2 Marks)
- (v) Standard deviation (2 Marks)
- b) PWC Kenya Limited recently employed some audit trainees. After one week, the supervisor noticed that the three trainees assigned to audit EABL accounts, Tom, David and Mercy had completed the assignment but he discovered that 2%, 3% and 5% of the accounts audited by Tom, David and Mercy respectively had errors. He had given Tom, David and Mercy to audit 40%, 35% and 25% of the accounts respectively

Determine the probability that:

- (i) An account with errors was audited by Mercy.
- (ii) Tom or David audited an account with no errors.
- c) Explain any five (5) random sampling methods.

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UNIVERSITY EXAMINATIONS 2020/2021

FIRST YEAR SECOND SEMESTER EXAMINATION FOR THE DEGREE OF
BACHELOR OF SCIENCE IN MATHEMATICS AND COMPUTER SCIENCE

SMA 2103: PROBABILITY AND STATISTICS I

DATE: 29TH JULY, 2021

TIME: 2 HOURS

INSTRUCTIONS:

ANSWER YOUR QUESTIONS IN THE ANSWER BOOKLET PROVIDED.
ANSWER QUESTION ONE AND ANY OTHER TWO QUESTIONS.

QUESTION ONE (THIRTY MARKS)

Differentiate between the two sources for data collection in Statistics. [2 Marks]

Let X be a random variable with $P(X = 1) = 0.2$, $P(X = 2) = 0.3$ and $P(X = 3) = 0.5$. What is the expected value and standard deviation of:

i). X [3 Marks]

ii). $Y = 5X - 10$ [3 Marks]

The following elements comprise the earth's crust. Illustrate and analyze the composition of this crust with a Pareto Chart for this data.

Element	Oxygen	Calcium	Aluminum	Iron	Silicon	Others
Percentage	45.6	4.7	8.4	6.2	27.3	7.8

[4 Marks]

A bag contains 7 green balls and 5 red balls. If two balls are drawn, what is the probability that the balls are:

i). Both red. [2 Marks]

ii). Of different colors. [2 Marks]

The summary statistics given below are from an analysis of data from a study conducted to establish whether variables X and Y are related: $\Sigma X = 185$, $\Sigma Y = 161$, $\Sigma XY = 2565$, $\Sigma X^2 = 3073$, $n = 12$. If variables X and Y are related, fit a simple linear regression equation for Y on X ($y = a + bx$). [4 Marks]

- [f]. i). Differentiate between the measures of skewness and kurtosis. [2 Marks]
 ii). The following figures represent the expenditure incurred by different families.
 Compute the Bowley's coefficient of skewness and interpret the results.

Expenditure	1-5	6-10	11-15	16-20	21-25	26-30	31-35
Families	29	53	48	70	38	27	20

[8Marks]

QUESTION TWO (TWENTY MARKS)

- [a]. Using examples for each, list down the two classifications of measures of location. [4Marks]
 [b]. Estimate the mean, median and mode of the following frequency distribution.

Class	10-14	15-19	20-24	25-29	30-34	35-39	40-44
Frequency	7	11	14	13	5	19	9

[9Marks]

- [c]. Find the lower and upper quartiles, the 7th decile and the 85th percentile of the following scores obtained by students: 27, 82, 52, 43, 31, 27, 79, 73, 66, 57 and 48. [7Marks]

QUESTION THREE (TWENTY MARKS)

- [a]. Identify and explain the three approaches to probability. [6Marks]
 [b]. At a certain assembly plant, three machines make 30%, 45% and 25% respectively, of the products. It is known from the past experience that 2%, 3% and 2% of the products made by each machine respectively is defective. Suppose a finished product is randomly selected.
 i). i) What is the probability that it is defective? [3 Marks]
 ii). ii) If a product is chosen randomly and found to be defective, what is the probability that it was made by machine 2? [2Marks]
 [c]. The table below shows the results obtained from a study conducted to determine whether variable X and Y are related.

X	78	86	72	82	80	86	84	89	68	71
Y	140	160	134	144	180	176	174	178	128	132

Determine the Pearson's Moment Correlation Coefficient and interpret your results.

[9Marks]

QUESTION FOUR (TWENTY MARKS)

[a]. i). Define the term sampling.

[2Marks]

ii). Identify and describe the five types of probability sampling techniques.

[10Marks]

[b]. The random variable Z has the probability distribution shown below:

z	2	3	5	7	11
$P(Z = z)$	$\frac{1}{6}$	$\frac{1}{3}$	$\frac{1}{4}$	x	y

If $E(Z) = 4\frac{2}{3}$, find the values of x and y hence determine the variance of Z. [8Marks]

QUESTION FIVE (TWENTY MARKS)

The following data represents the weekly wages of 20 employees at a production company.

Wages (\$)	340-349	350-359	360-369	370-379	380-389
Employees	2	3	7	5	3

Calculate:

- The mean and standard deviation using the coding formulae.
- Kelly's coefficient of skewness and interpret the results.
- The Inter-Quartile range.

[8Marks]

[7Marks]

[5Marks]

1. A box contains 10 balls numbered 1 to 10. A ball is drawn at random.

(a) List the sample space.

(b) List the event that the number on the ball is even.

2. A box contains 10 balls numbered 1 to 10. A ball is drawn at random.

(a) List the sample space.

(b) List the event that the number on the ball is even.

3. A box contains 10 balls numbered 1 to 10. A ball is drawn at random.

4. A box contains 10 balls numbered 1 to 10. A ball is drawn at random.

5. A box contains 10 balls numbered 1 to 10. A ball is drawn at random.

6. A box contains 10 balls numbered 1 to 10. A ball is drawn at random.

7. A box contains 10 balls numbered 1 to 10. A ball is drawn at random.

8. A box contains 10 balls numbered 1 to 10. A ball is drawn at random.

9. A box contains 10 balls numbered 1 to 10. A ball is drawn at random.

10. A box contains 10 balls numbered 1 to 10. A ball is drawn at random.

(2 marks)

(2 marks)

(2 marks)

11. The events A and B are such that $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{4}$ and $P(A \cap B) = \frac{1}{8}$.

(2 marks)

(1 marks)

(2 marks)

12. The random variable X has the following distribution.

X	1	2
$P(X = x)$	a	b

where a and b are constants.

(a) Write down $P(X)$.

(2 marks)

(2 marks)

(b) Given that $P(X) = 0.75$, find the values of a and b .

- e) The discrete random variable X can only take the values 1, 2, 3 and 4. For these values the cumulative distribution function is defined by

$$F(x) = kx^2 \quad \text{for } x = 1, 2, 3, 4; \quad \text{where } k \text{ is a constant.}$$

- i) Find the value of k . (2marks)

- ii) Find the probability distribution of X (2marks)

- f) Given the following summary data,

$$\sum x = 367; \sum y = 270; \sum x^2 = 33845; \sum y^2 = 12976; \sum xy = 17135; n = 6$$

Calculate the product moment correlation coefficient (r) using the formula:

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

(4marks)

- g) The following stem and leaf diagram shows the population scores, x , obtained by a group of students in a test.

Score	
2	1 2 8 (3)
3	3 4 7 8 9 (5)
4	1 2 3 5 6 7 9 (7)
5	0 2 3 3 5 5 6 8 9 9 (10)
6	1 2 2 3 4 4 5 6 6 8 8 9 9 (14)
7	0 2 3 4 5 7 8 9 (8)
8	0 1 4 (3)

Key: 6 | 1 means 61

- (i) State the population median score. (1mark)

- (ii) Given further that for these data: $\sum x = 2873$ and $\sum x^2 = 177353$, calculate, to two decimal places, the mean and the standard deviation for these data. (3marks)

- (iii) Use the formula: $\frac{3(\text{mean} - \text{median})}{\text{standard deviation}}$ to show that these data are negatively skewed. (1mark)

- h) What is data sampling? (1mark)

- i) State one advantage of data sampling. (1mark)

a) The discrete random variable X has a probability function:

$$P(X = x) = \begin{cases} kx, & x = 1, 2, 3, 4, 5. \\ 0, & \text{Otherwise} \end{cases}$$

Show that $k = \frac{1}{15}$.

b) Find the value, from (a) above, of

i) $E(2X + 3)$

ii) $Var(2X - 4)$

c) State the Bayes' theorem.

d) For any married couple who are members of a tennis club, the probability that the husband has a degree is $\frac{3}{5}$, and the probability that the wife has a degree is $\frac{1}{2}$. The probability that the

husband has a degree given that the wife has a degree is $\frac{11}{12}$. A married couple is chosen at

random. Show that the probability that both of them have degrees is $\frac{11}{24}$.

e) Draw a Venn diagram to represent the data in (d) above

f) Find the probability that:

i) only one of them has a degree

ii) neither of them has a degree

g) Two married couples are chosen at random. Find the probability that only one of the two

husbands and only one of the two wives have degrees.

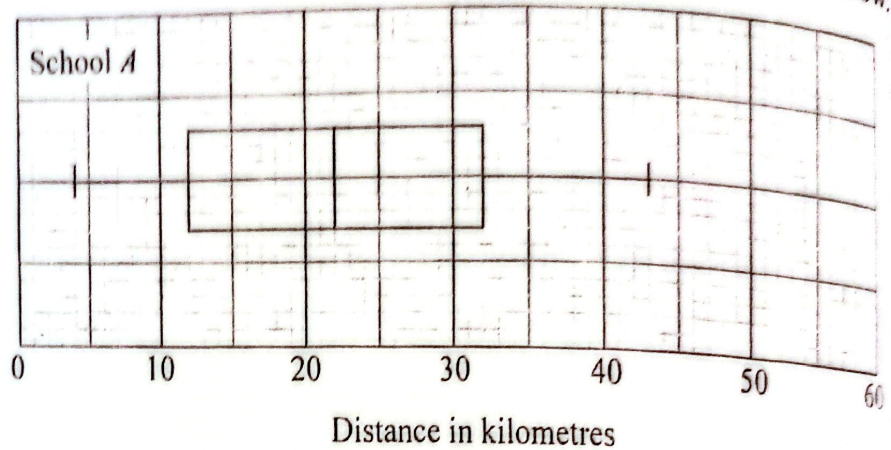
QUESTION THREE (TWENTY MARKS)

a) A class of 16 boys and 13 girls did a Physics test. The test was marked out of 60. Their marks are shown below.

Boys	Girls
45	26
54	54
32	47
28	34
34	52
54	24
56	45
48	34
32	52
29	24
47	45
48	34
44	52
45	24
56	45
57	34

i) Draw a back-to-back stem and leaf diagram to represent these data.

- ii) Comment on your results. (1mark)
- b) State two advantages and two disadvantages of using the quartiles to summarize a set of data. (4marks)
- c) State any two features and one use of a box plot. (3 marks)
- d) The distances in, kilometers, travelled to school by the teachers in two schools A and B, in the same town were recorded. The data for school A are summarized in the diagram below.



- e) For school B, the least distance travelled was 3 km and the longest distance travelled was 55 km. The three quartiles were 17, 24 and 31 respectively. An outlier is an observation that falls either $1.5 \times (\text{Interquartile range})$ above the upper quartile or $1.5 \times (\text{Interquartile range})$ below the lower quartile.
- i) Draw a box plot for school B (5marks)
- ii) Compare and contrast the two box plots. (4marks)

QUESTION FOUR (TWENTY MARKS)

- a) In the study of a city, the population density, in people/hectare, and the distance from the city centre, in km, was investigated by picking a number of sample areas with the following results.

Area	A	B	C	D	E	F	G	H	I	J
Distance (km)	0.6	3.8	2.4	3.0	2.0	1.5	1.8	3.4	4.0	0.9
Population Density (people/hectare)	50	22	14	20	33	47	25	8	16	38

- i) Plot a scatter diagram. (4marks)
- ii) Describe the type of correlation. (1mark)
- iii) Interpret the correlation. (1mark)

- b) A drilling machine can run at various speeds, but in general the higher the speed the sooner the drill needs to be replaced. Over several months, 15 pairs of observations relating to speed, s revolutions per minute, and life of drill, h hours, are collected.
- For convenience the data are coded so that $x = s - 20$ and $y = h - 100$ and the following summations obtained.

$$\sum x = 143; \sum y = 391; \sum x^2 = 2413; \sum y^2 = 22441; \sum xy = 484.$$

- Find the equation of the regression line of h on s . (10marks)
- Interpret the slope of your regression line. (2marks)
- Estimate the life of a drill revolving at 30 revolutions per minute. (2marks)

QUESTION FIVE (TWENTY MARKS)

- Give the difference between a bar graph and a histogram as methods of data presentation. (2marks)
- The following table summarizes the distances, to the nearest kilometer, that 134 examiners travelled to attend a meeting in London.

Distance (km)	Number of examiners
41 - 45	4
46 - 50	19
51 - 60	53
61 - 70	37
71 - 90	15
91 - 150	6

- Give a reason to justify the use of a histogram to represent these data. (1mark)
- Calculate the frequency densities needed to draw a histogram for these data. (Do not draw the histogram) (3marks)
- Use interpolation to estimate the median Q_2 , the lower quartile, Q_1 and the upper quartile Q_3 of these data. (6marks)
- The midpoint of each class is represented by x and the corresponding frequency by f . Calculations then give the following values:

$$\sum fx = 8379.5 \text{ and } \sum fx^2 = 557489.75$$

Use the given information to calculate an estimate (for these data) of:

- the mean.

(ii) the standard deviation.

(3marks)

f) One coefficient of skewness is given by,

$$\frac{Q_3 - 2Q_2 + Q_1}{Q_3 - Q_1}$$

(i) Evaluate this coefficient and comment on the skewness of these data.

(2marks)

(ii) Give another justification of your comment in f (i) above.

(1mark)

